

WELLGUARD AI

Intelligent Predictive Maintenance Report Electrical Submersible Pumps (ESP)

Well Identifier:	B-167
Analysis Date:	2026-07-16 04:37:43
Primary Diagnoses:	Warning (Warning)
Remaining Useful Life:	18 Days

CONFIDENTIAL NOTICE: This document contains proprietary analysis for oil field operations.

1. Executive Summary

Based on real-time sensor analytics, Well **B-167** is operating in a **WARNING** state. The computed Pump Health Index (PHI) is currently **56.0%**. Our AI models forecast that the pump has approximately **18 days** of Remaining Useful Life before maintenance intervention is mandatory.

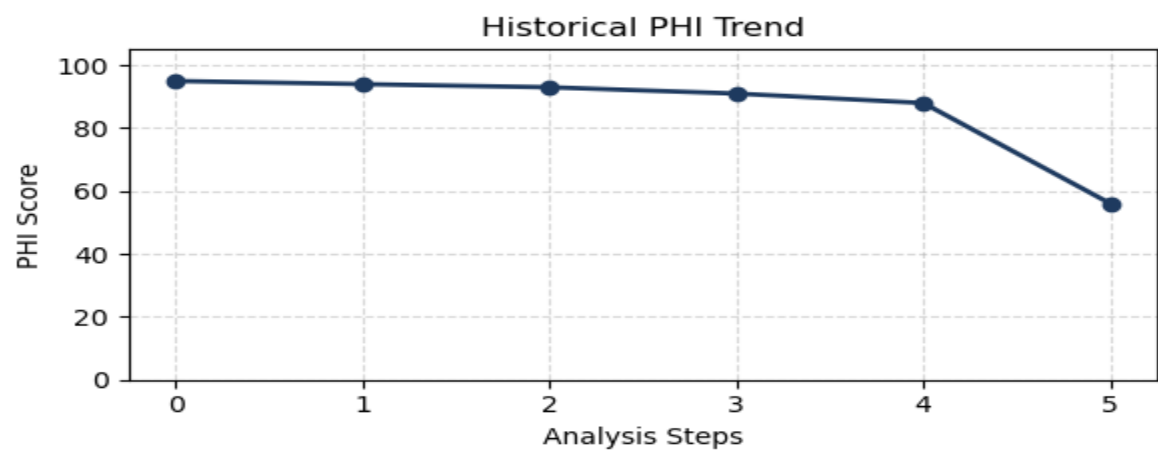
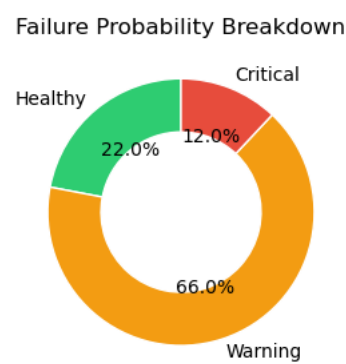
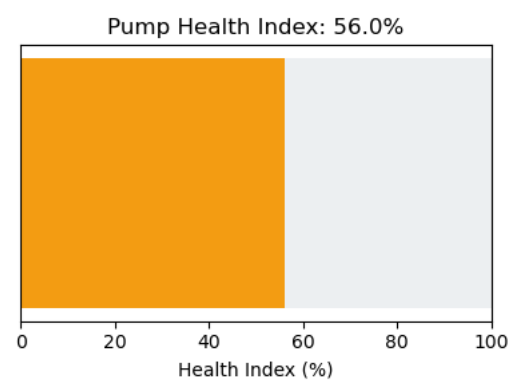
Primary Recommendation:

Action Required: Telemetry Sensor Freeze

Details: Downhole sensor signals (discharge/intake pressure, temperature) are completely frozen. Telemetry failure detected.

Suggested Response Time: Immediate

2. Health Dashboard Visualizations



3. Real-time Sensor Values Analysis

Sensor Channel	Current Reading	Baseline Range	Hourly Trend
Motor Temperature	174.7	210.0 °F	→ Stable
Motor Current	32.0	32.3 A	→ Stable
Discharge Pressure	944.4	2395.8 psi	→ Stable
Intake Pressure	925.1	837.8 psi	→ Stable
Hours_Since_Restart	240.0	N/A	→ Stable

4. Key AI Findings & Anomaly Detection

Our predictive maintenance models have analyzed current sensor anomalies and temporal progressions. Here are the specific patterns identified:

- **Frozen Signal Fault:** Sensor values indicate a flatline condition. The downhole instruments are frozen or telemetry link is lost.

5. Detailed Maintenance Recommendations

Priority	Title & Recommended Maintenance Action	Timeline
P1 - Urgent	Telemetry Sensor Freeze: Downhole sensor signals (discharge/intake pressure, temperature) are completely frozen. Telemetry failure detected.	Immediate

Appendix: Glossary & Confidence Levels

Term	Definition
ESP	Electric Submersible Pump
PHI	Pump Health Index (0-100% conditional ranking)
RUL	Remaining Useful Life (estimated active run days)
VSD/VFD	Variable Speed/Frequency Drive controller

Model Confidence Metric: Best Selected Classifier (Random Forest) validation Macro F1 score: **1.0000**